



US 20250280922A1

(19) **United States**

(12) **Patent Application Publication**
OLESON et al.

(10) **Pub. No.: US 2025/0280922 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **SEGMENTED SOLE FOR FOOTWEAR**

Publication Classification

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(51) **Int. Cl.**
A43B 13/14 (2006.01)
A43B 5/06 (2022.01)
A43B 13/04 (2006.01)
A43B 13/12 (2006.01)
A43D 27/02 (2006.01)

(52) **U.S. Cl.**
CPC *A43B 13/141* (2013.01); *A43B 5/06*
(2013.01); *A43B 13/04* (2013.01); *A43B*
13/122 (2013.01); *A43B 13/148* (2013.01);
A43D 27/02 (2013.01)

(21) Appl. No.: **19/216,412**

(57) **ABSTRACT**

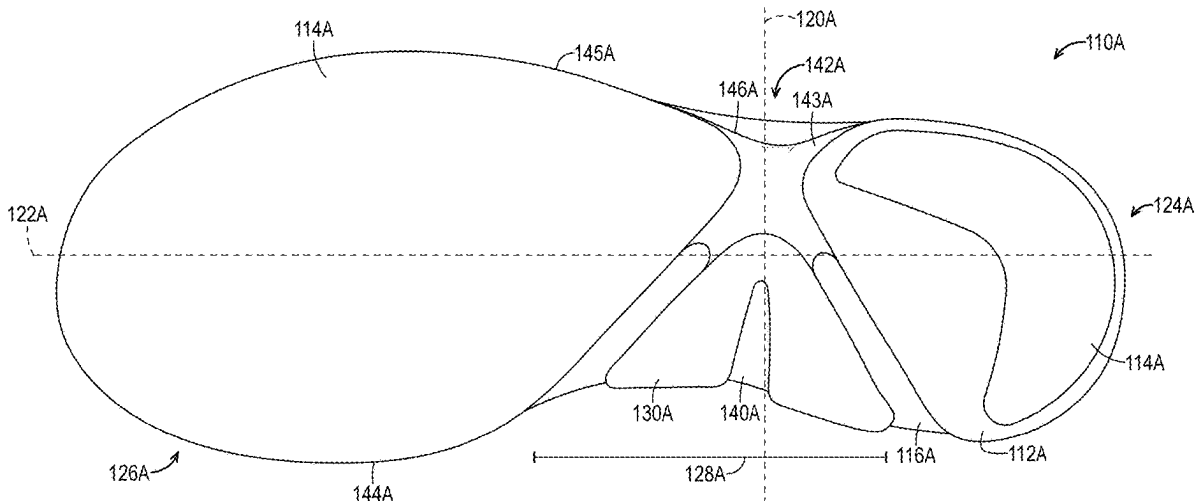
(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/639,235, filed on
Feb. 28, 2022, filed as application No. PCT/CA2020/
051183 on Aug. 28, 2020.

(60) Provisional application No. 62/894,653, filed on Aug.
30, 2019.

A segmented sole for an article of footwear comprises a forefoot region, a heel region, and a segmentation zone disposed between the forefoot and heel region. The segmentation zone includes a plurality of segments separated from each other and from the forefoot and heel by a plurality of sipes, cuts, or gaps. The segmented sole produces bending along a segmentation axis disposed along an arch of a wearer of an article of footwear including the sole, allowing for arch support and natural foot movement.



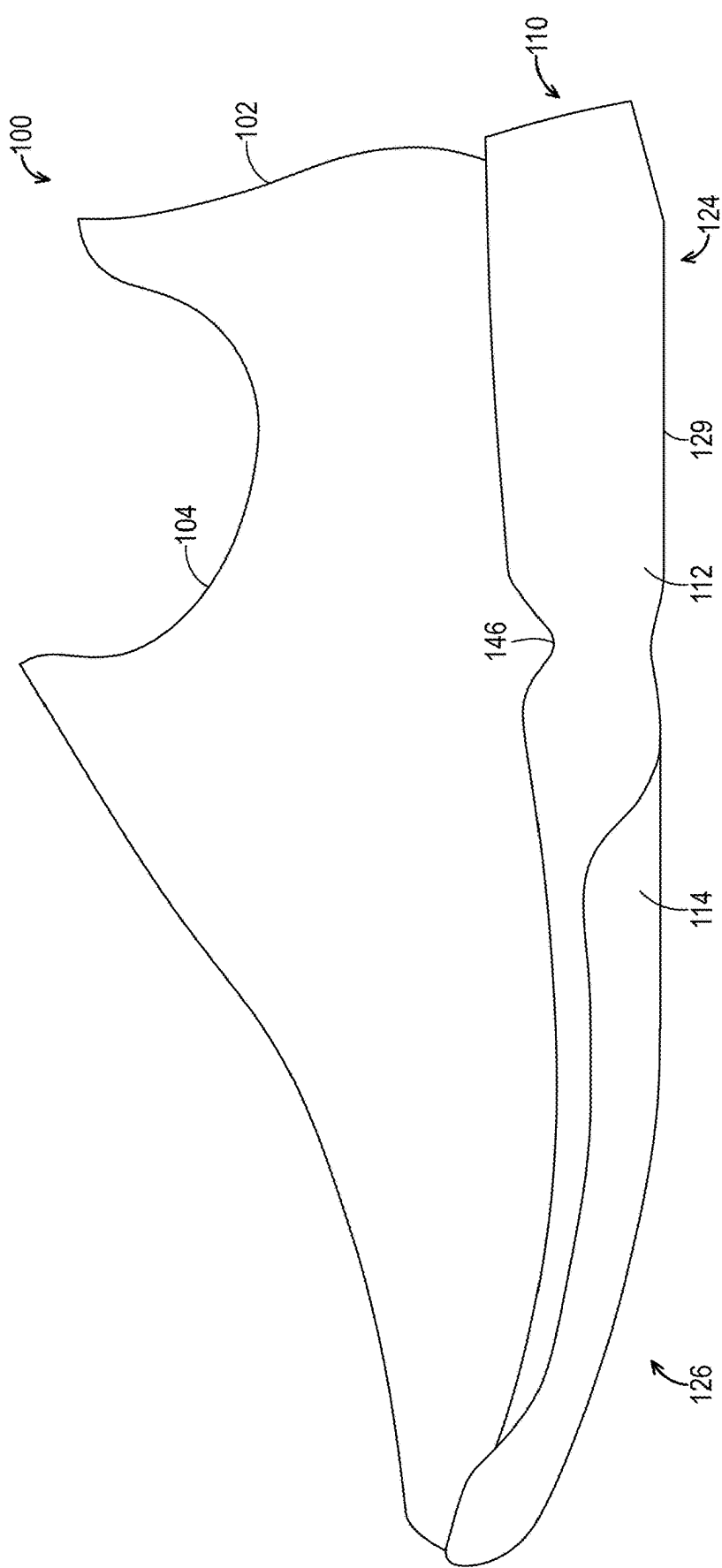


FIG. 1

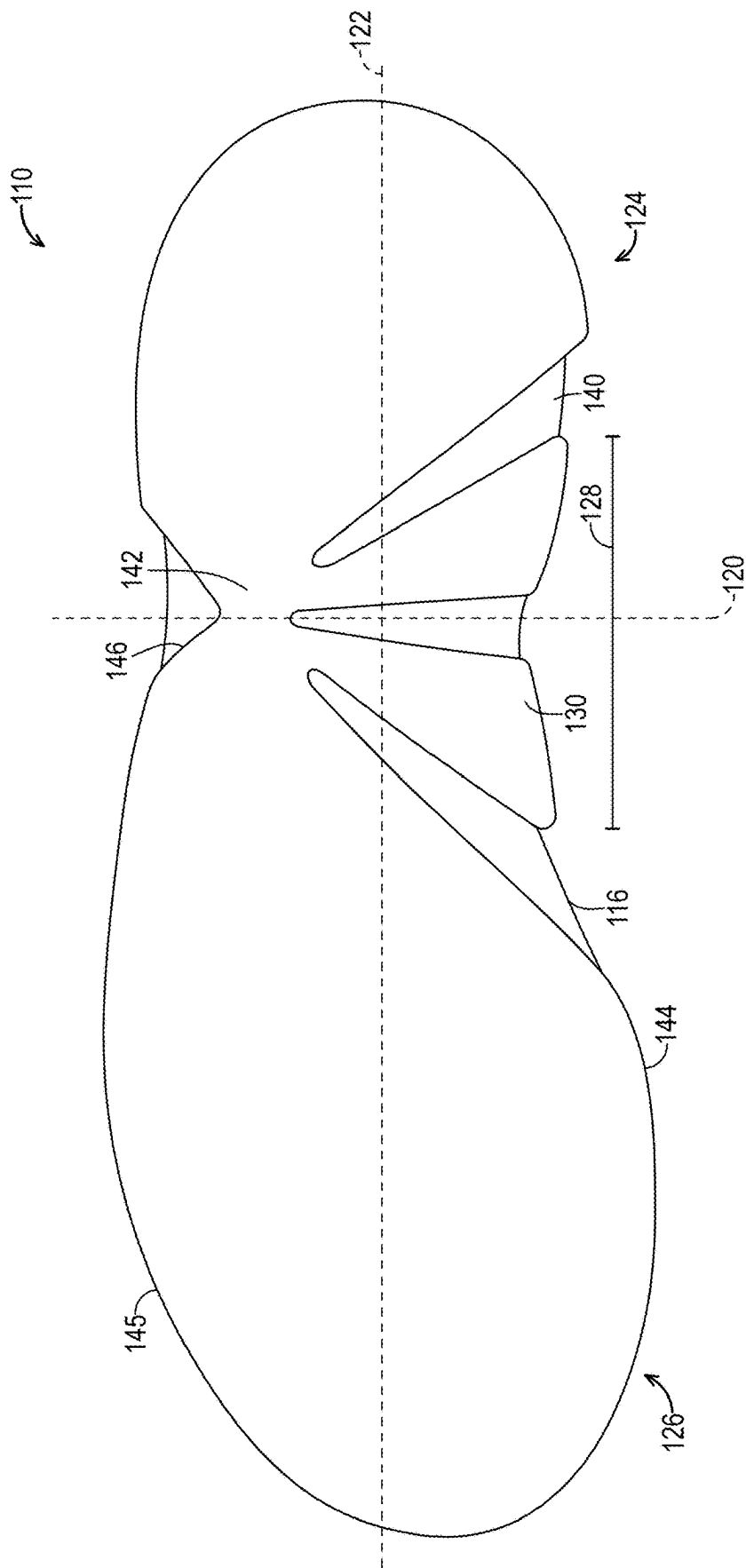


FIG. 2

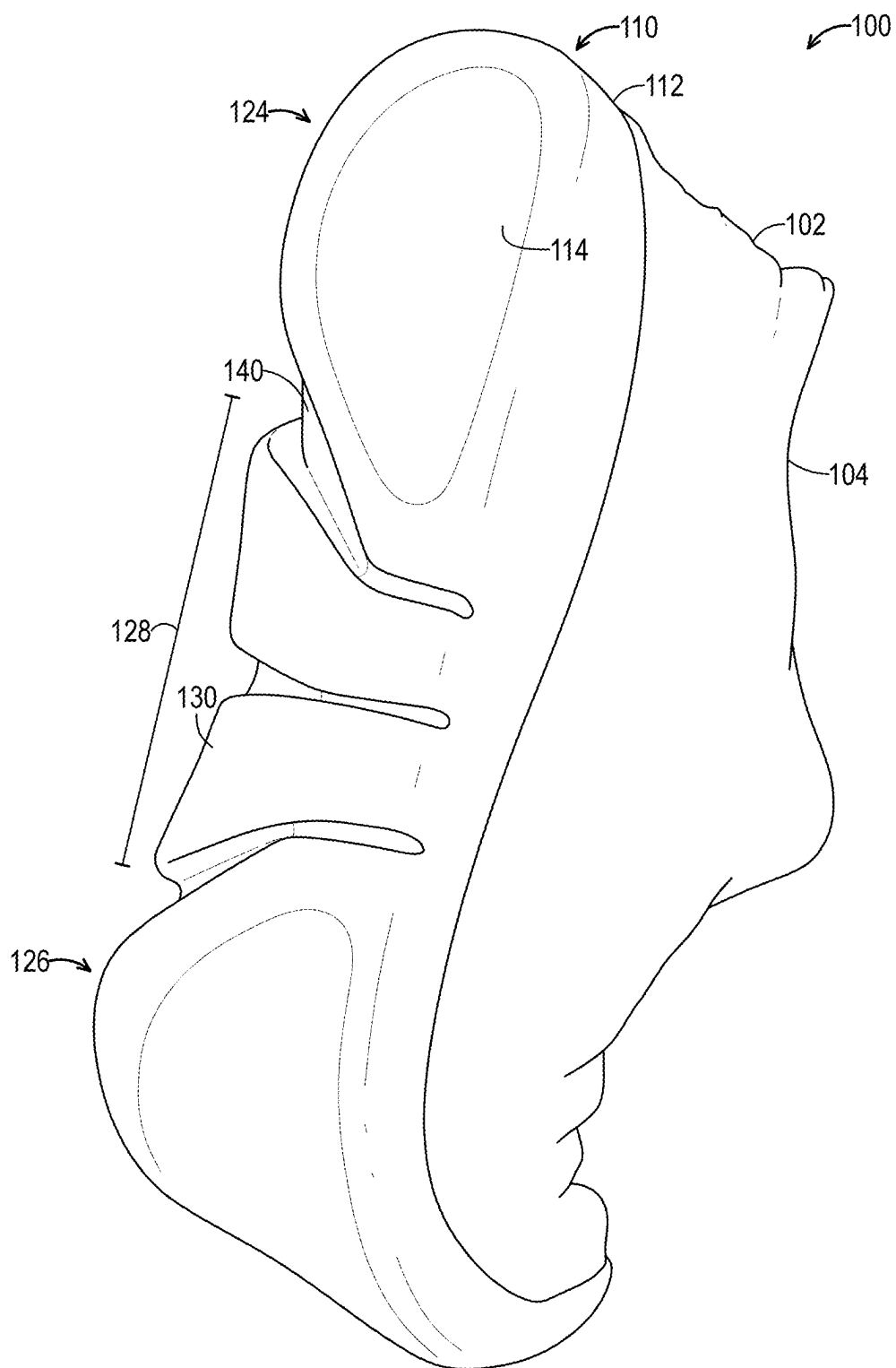


FIG. 3

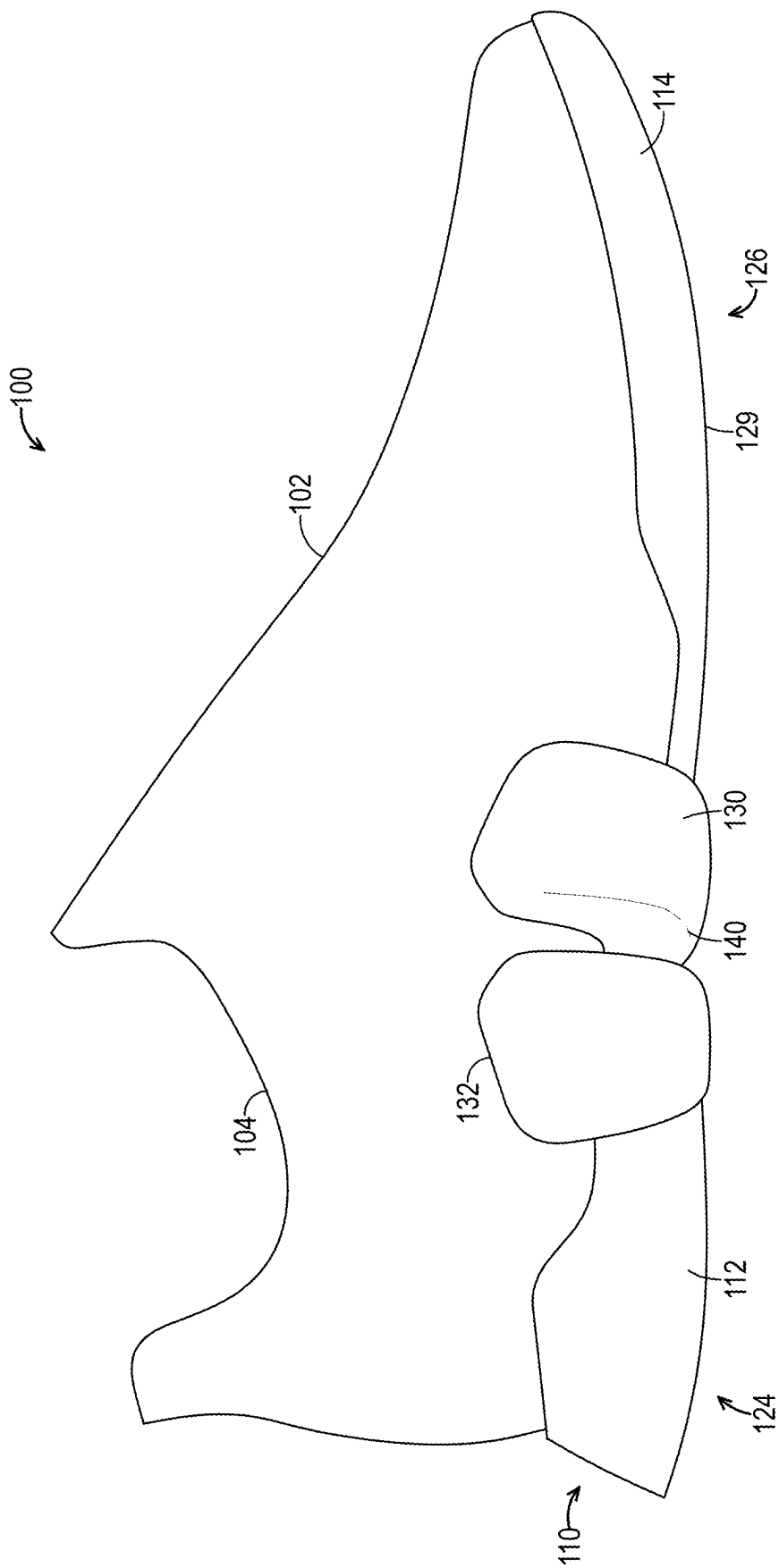


FIG. 4

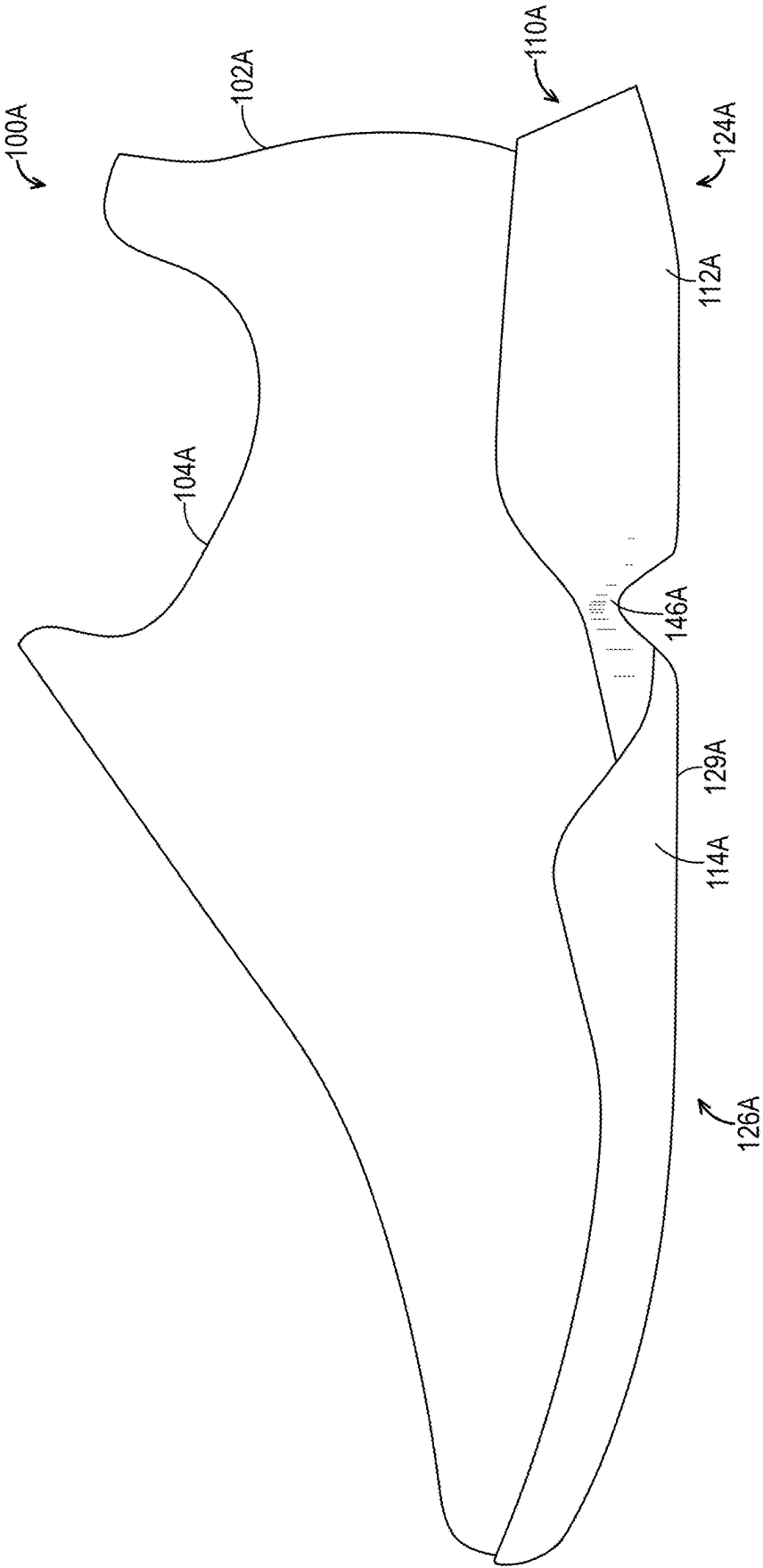


FIG. 5

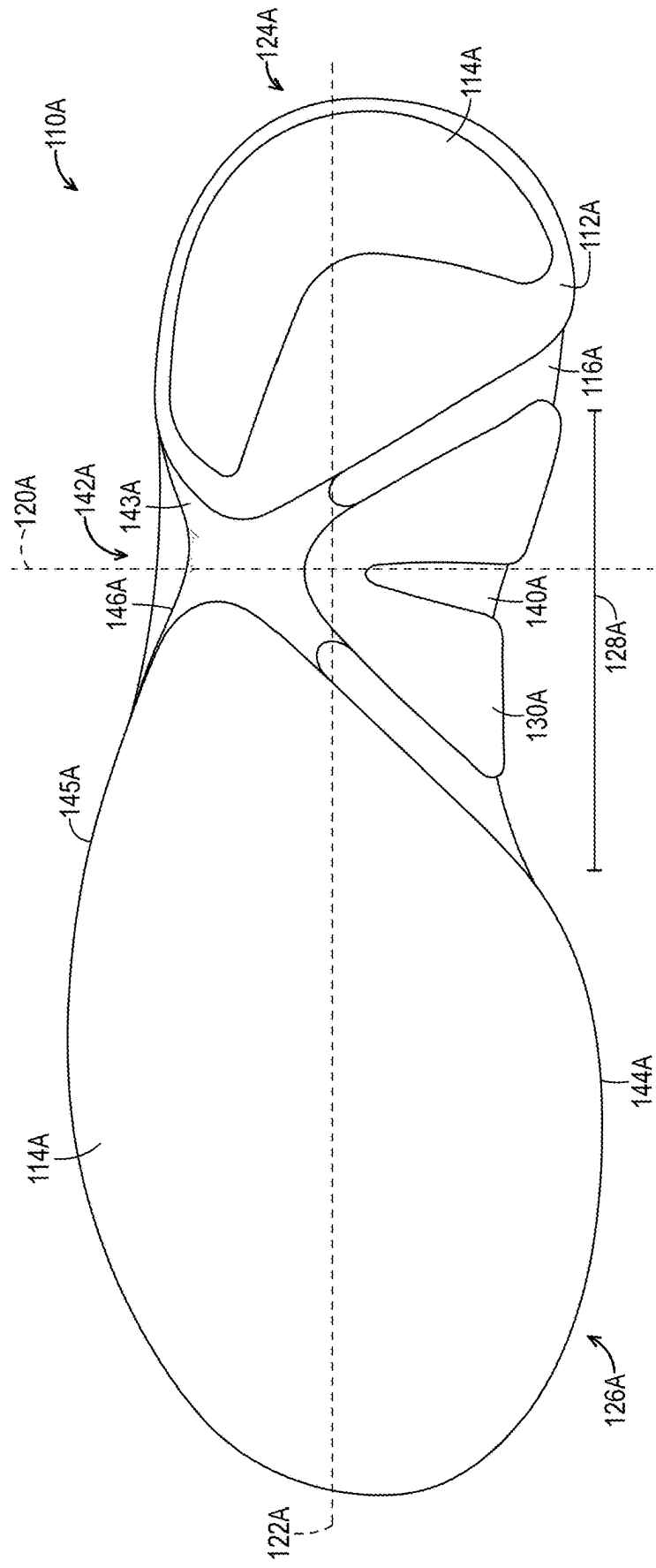


FIG. 6

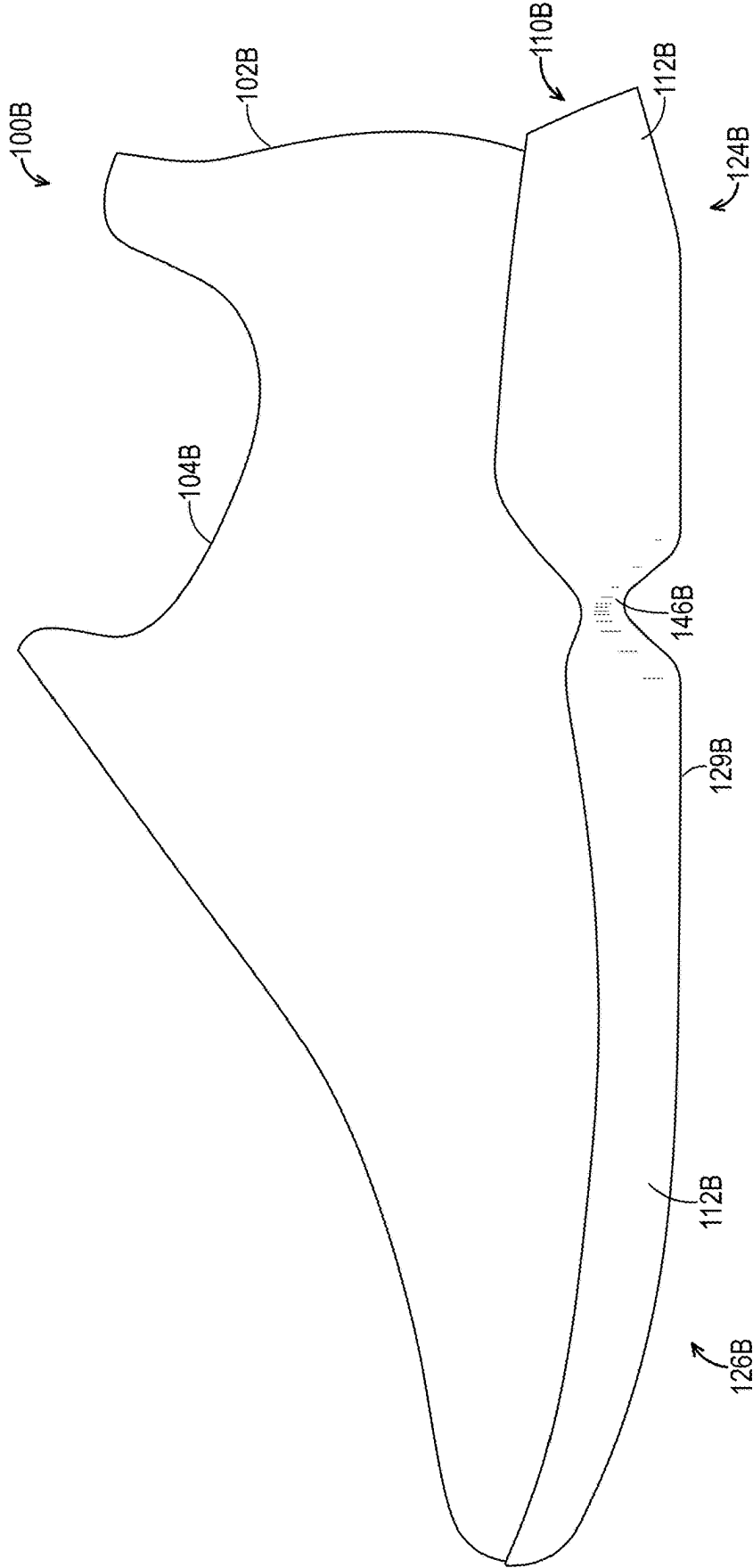


FIG. 7

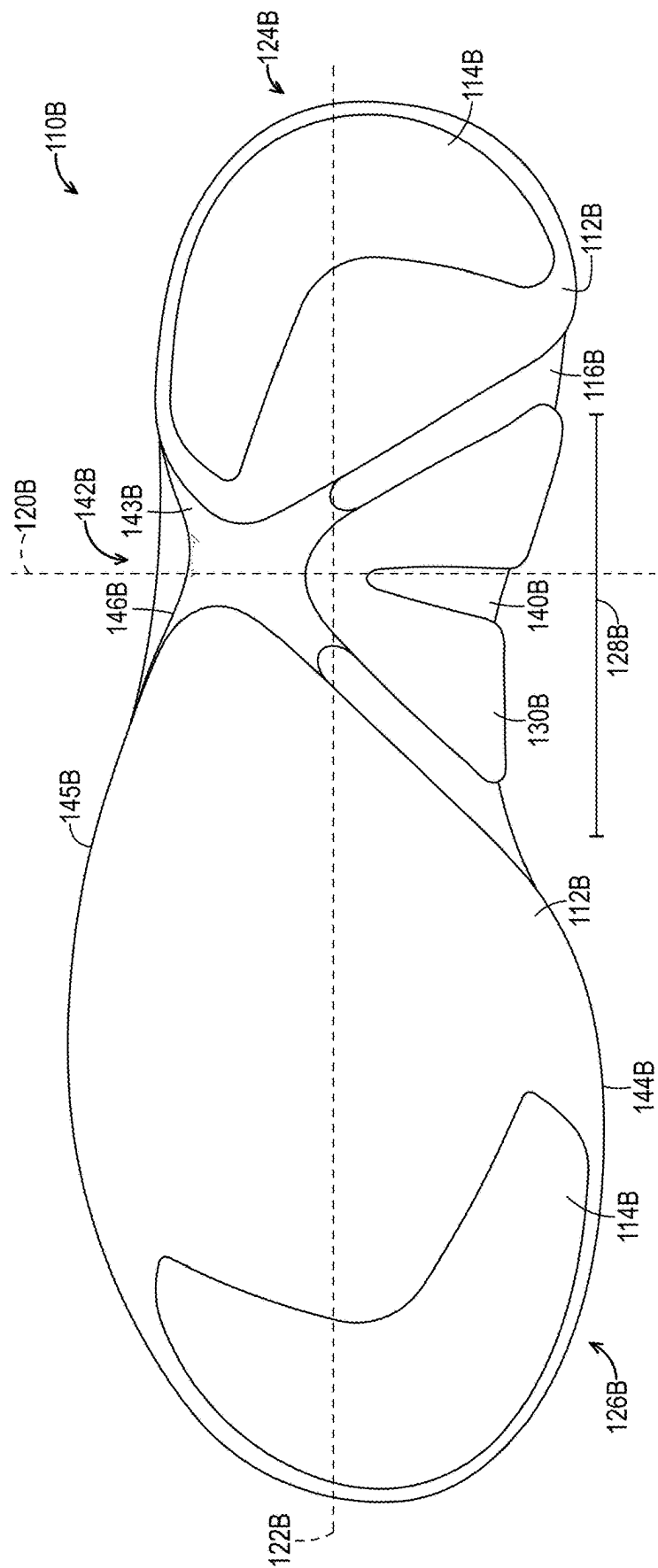


FIG. 8

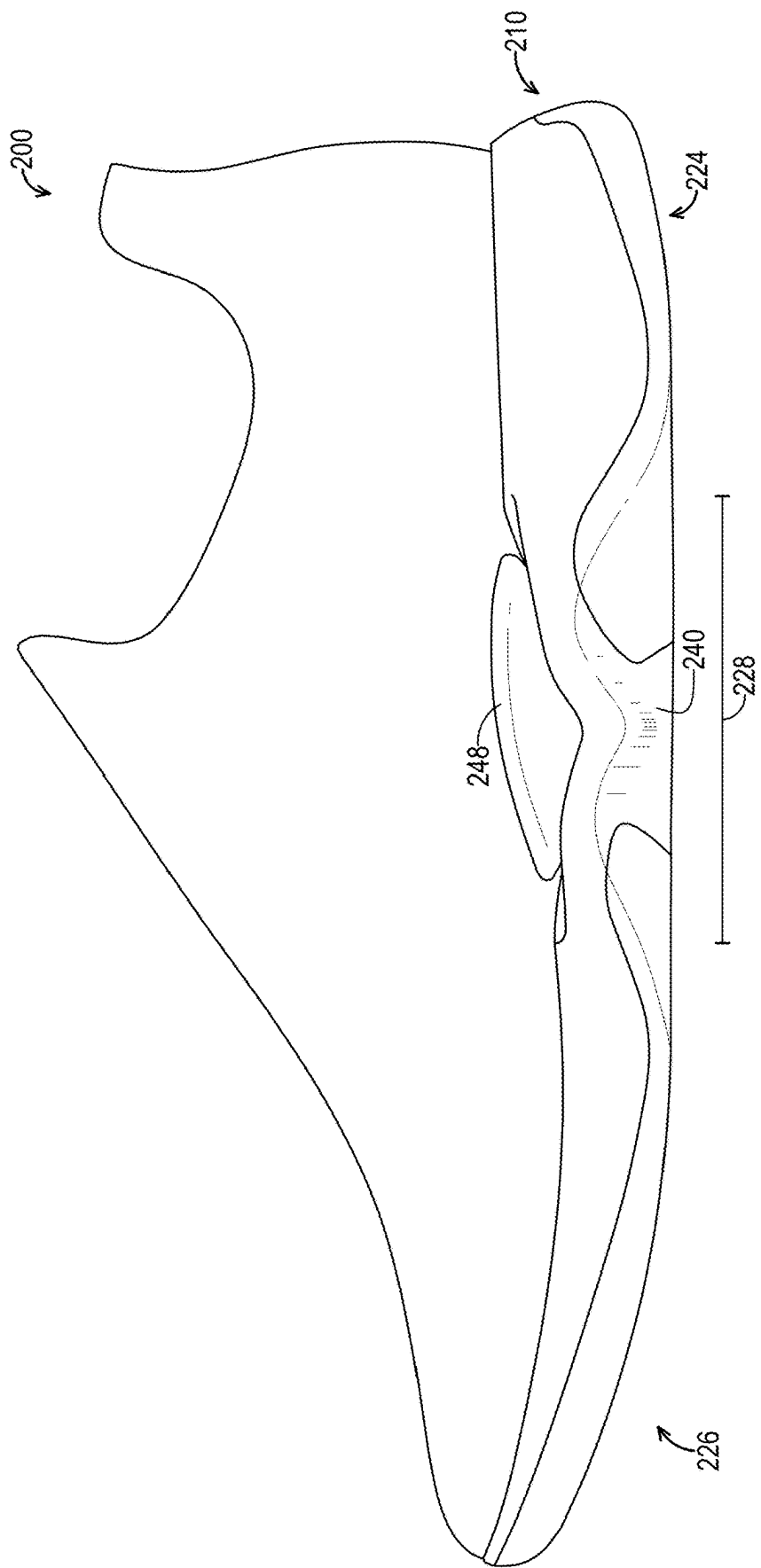


FIG. 9

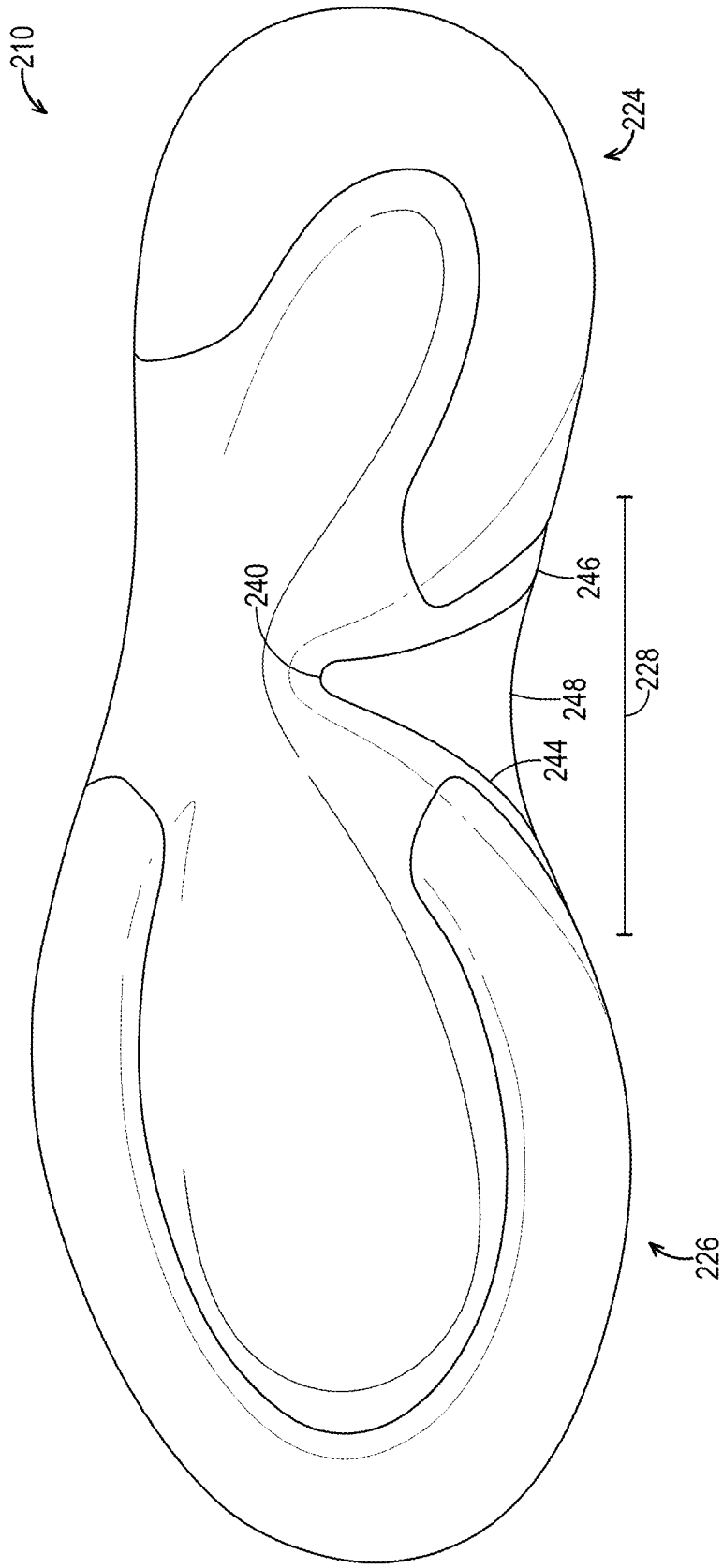


FIG. 10

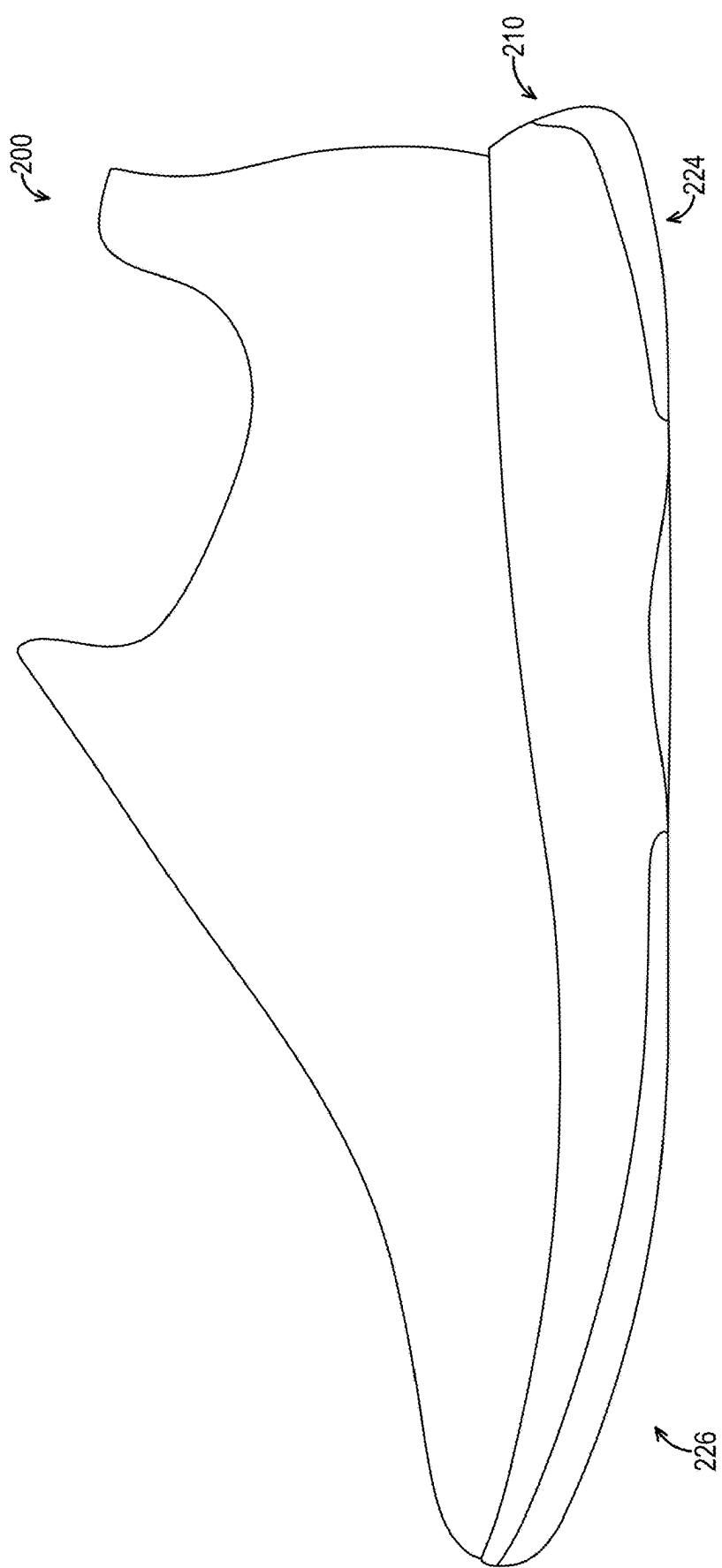


FIG. 11

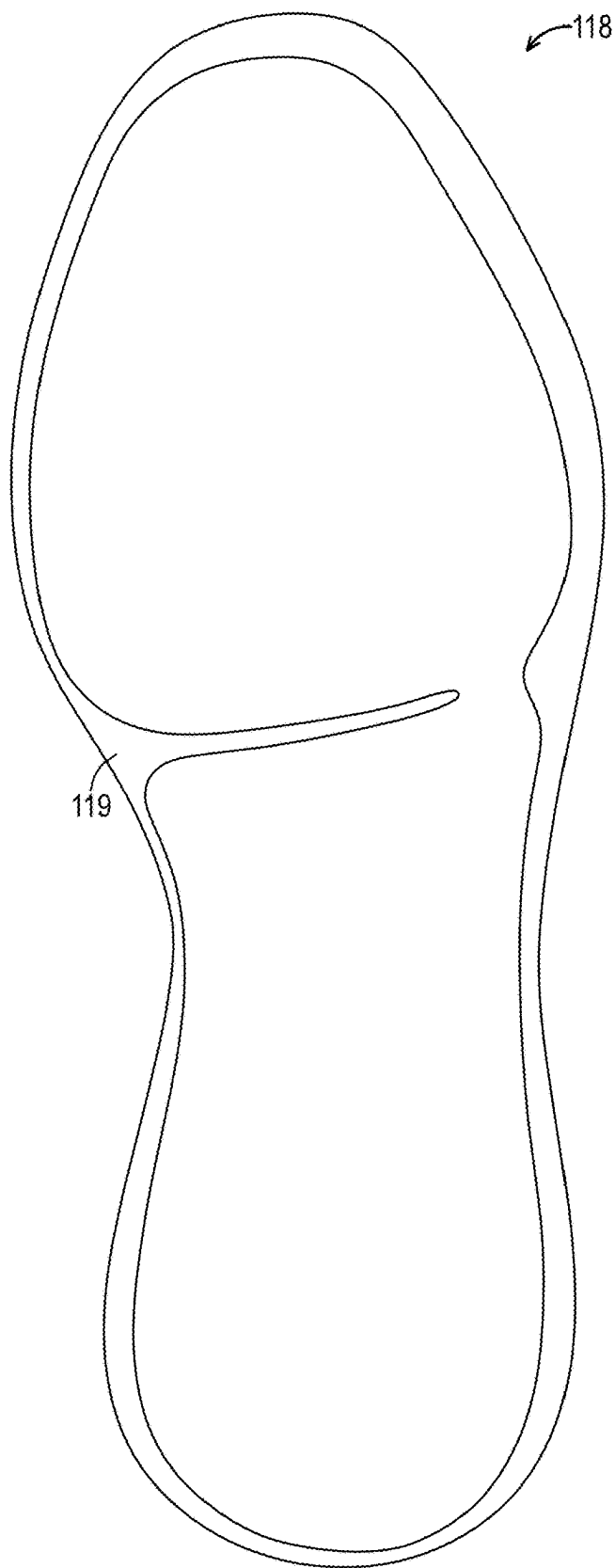


FIG. 12

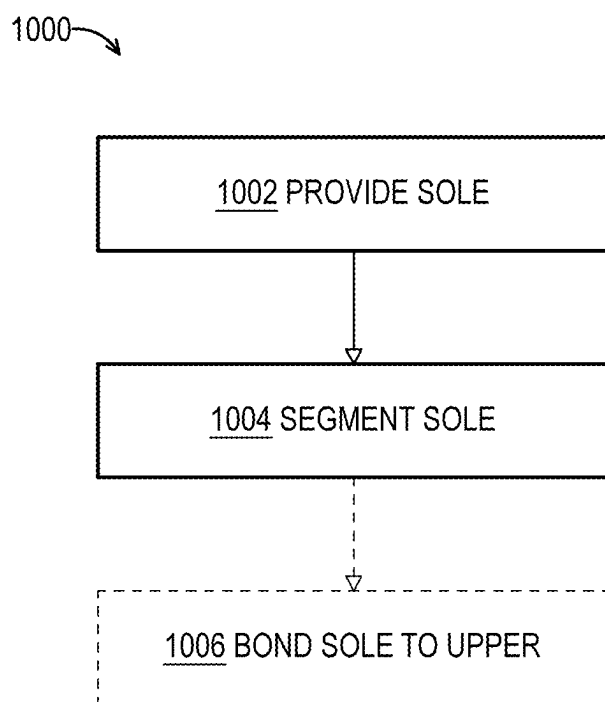


FIG. 13

SEGMENTED SOLE FOR FOOTWEAR

CROSS-REFERENCES

[0001] The following applications and materials are incorporated herein, in their entireties, for all purposes: U.S. Provisional Patent Application Ser. No. 62/894,653, filed Aug. 30, 2019; International Patent Application Serial No. PCT/CA2020/051183, filed Aug. 28, 2020; and U.S. patent application Ser. No. 17/639,235, filed Feb. 28, 2022.

FIELD

[0002] This disclosure relates to systems and methods for footwear. More specifically, the disclosed embodiments relate to flexible footwear soles.

INTRODUCTION

[0003] Athletic shoes are generally designed to provide some level of support to a user's foot, while also allowing some level of free movement of the foot itself. Shoes may have various levels of flexibility and support based on an intended purpose of the shoe and user's preferences. Increasing shoe flexibility often decreases shoe supportiveness and vice versa. Shoe designers and manufacturers must balance these trade-offs to provide shoes which have preferred levels of flexibility and support.

[0004] Minimalist running shoes, for example, provide low levels of support to a wearer, are popular with runners and gym-goers who appreciate freedom of movement and a "barefoot" feeling while running. Minimalist running shoes may provide low levels of arch support and lightweight soles, allowing a wearer to experience greater contact with the ground. These shoes may provide adequate support for some runners, but may not provide adequate support for runners who are injury-prone or who consistently run on hard surfaces.

[0005] Minimalist shoes may be contrasted with conventional running shoes, which provide high levels of stability and support, preventing overpronation and absorbing shock associated with the foot striking the ground. However, conventional running shoes may be too heavy for some runners and may cause certain foot muscles to be underdeveloped. Conventional running shoes may also be unsuitable for gym workouts and other forms of exercise, which require greater flexibility along the arch of a wearer. Some athletes may appreciate the benefits of running or training shoes which provide flexibility while also providing arch support and shock absorption.

[0006] There is a need for shoes which accommodate anatomical movement within the feet of a wearer, while still preventing injuries which may be associated with rolled ankles or repeated shock from feet striking the ground.

SUMMARY

[0007] The present disclosure provides systems, apparatuses, and methods relating to segmented soles for footwear.

[0008] In some embodiments, a sole for an article of footwear may include: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a first sipe on a medial edge of the sole and a second sipe on the medial edge of the sole, wherein the first sipe and the second sipe angle toward each other, intersect at a convergence point inset from a lateral edge of the sole, and combine into

a combined sipe extending between the convergence point and the lateral edge of the sole.

[0009] In some embodiments, a sole for an article of footwear may include: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a first sipe extending between a medial edge of the sole and a lateral edge of the sole and a second sipe extending between the medial edge of the sole and the lateral edge of the sole, wherein the first sipe and the second sipe intersect at a convergence point inset from the lateral edge of the sole and define a triangular segment between the medial edge of the sole and the convergence point.

[0010] In some embodiments, a sole for an article of footwear may include: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a first sipe extending between a medial edge of the sole and a lateral edge of the sole, a second sipe extending between the medial edge of the sole and the lateral edge of the sole, and a third sipe, wherein the first sipe and the second sipe meet defining a segment between the medial edge and the lateral edge, and the third sipe extends into and partially divides the segment.

[0011] Features, functions, and advantages may be achieved independently in various embodiments of the present disclosure, or may be combined in yet other embodiments, further details of which can be seen with reference to the following description and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a left elevation view of a first illustrative article of footwear having a segmented sole.

[0013] FIG. 2 is a bottom plan view of the illustrative article of footwear of FIG. 1.

[0014] FIG. 3 is an isometric view of an illustrative article of footwear in a flexed position.

[0015] FIG. 4 is a right elevation view of the illustrative article of footwear of FIG. 1.

[0016] FIG. 5 is a left elevation view of an illustrative article of footwear having a segmented sole and a first outsole material ratio.

[0017] FIG. 6 is a bottom plan view of the illustrative article of footwear of FIG. 5.

[0018] FIG. 7 is a left elevation view of an illustrative article of footwear having a second outsole material ratio.

[0019] FIG. 8 is a bottom plan view of the illustrative article of footwear of FIG. 7.

[0020] FIG. 9 is a right elevation view of a second illustrative article of footwear having a segmented and flexible sole.

[0021] FIG. 10 is a bottom plan view of the second illustrative article of footwear of FIG. 9.

[0022] FIG. 11 is a left elevation view of the second illustrative article of footwear of FIG. 9.

[0023] FIG. 12 is a bottom plan view of an illustrative segmented insole suitable for use within the illustrative articles of footwear depicted in FIGS. 1-9.

[0024] FIG. 13 is a flowchart depicting steps of an illustrative method for manufacturing a segmented sole for an article of footwear in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0025] Various aspects and examples of segmented soles for footwear, as well as related methods, are described below and illustrated in the associated drawings. Unless otherwise specified, an outsole in accordance with the present teachings, and/or its various components, may contain at least one of the structures, components, functionalities, and/or variations described, illustrated, and/or incorporated herein. Furthermore, unless specifically excluded, the process steps, structures, components, functionalities, and/or variations described, illustrated, and/or incorporated herein in connection with the present teachings may be included in other similar devices and methods, including being interchangeable between disclosed embodiments. The following description of various examples is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. Additionally, the advantages provided by the examples and embodiments described below are illustrative in nature and not all examples and embodiments provide the same advantages or the same degree of advantages.

[0026] This Detailed Description includes the following sections, which follow immediately below: (1) Definitions; (2) Overview; (3) Examples, Components, and Alternatives; (4) Advantages, Features, and Benefits; and (5) Conclusion. The Examples, Components, and Alternatives section is further divided into subsections, each of which is labeled accordingly.

Definitions

[0027] The following definitions apply herein, unless otherwise indicated.

[0028] “Comprising,” “including,” and “having” (and conjugations thereof) are used interchangeably to mean including but not necessarily limited to, and are open-ended terms not intended to exclude additional, unrecited elements or method steps.

[0029] Terms such as “first,” “second,” and “third” are used to distinguish or identify various members of a group, or the like, and are not intended to show serial or numerical limitation.

[0030] “AKA” means “also known as,” and may be used to indicate an alternative or corresponding term for a given element or elements.

[0031] The terms “medial,” “lateral,” “anterior,” “posterior,” and the like are intended to refer to anatomical directions corresponding to a human on whom the article or object is configured to be placed or worn. For example, “medial” refers to a relative position disposed toward the center of the human body, while “lateral” refers to a relative position disposed away from the center of the human body. With respect to footwear, the term “anterior” refers to a relative position closer to the toe of a wearer and “posterior” refers to a relative position closer to the heel of the wearer. In the absence of a wearer, the same directional terms may be used as if the article of footwear is being worn in its expected configuration.

[0032] “Elongate” or “elongated” refers to an object or aperture that has a length greater than its own width, although the width need not be uniform. For example, an elongate slot may be elliptical or stadium-shaped, and an elongate candlestick may have a height greater than its

tapering diameter. As a negative example, a circular aperture would not be considered an elongate aperture.

[0033] “Coupled” means connected, either permanently or releasably, whether directly or indirectly through intervening components.

[0034] “Resilient” describes a material or structure configured to respond to normal operating loads (e.g., when compressed) by deforming elastically and returning to an original shape or position when unloaded.

[0035] “Rigid” describes a material or structure configured to be stiff, non-deformable, or substantially lacking in flexibility under normal operating conditions.

[0036] “Elastic” describes a material or structure configured to spontaneously resume its former shape after being stretched or expanded.

[0037] Directional terms such as “up,” “down,” “vertical,” “horizontal,” and the like should be understood in the context of the particular object in question. For example, an object may be oriented around defined X, Y, and Z axes. In those examples, the X-Y plane will define horizontal, with up being defined as the positive Z direction and down being defined as the negative Z direction.

[0038] “Providing,” in the context of a method, may include receiving, obtaining, purchasing, manufacturing, generating, processing, preprocessing, and/or the like, such that the object or material provided is in a state and configuration for other steps to be carried out.

[0039] “EVA” refers to ethylene-vinyl acetate.

[0040] “TPU” refers to thermoplastic polyurethane.

[0041] “PU” refers to polyurethane.

[0042] “TPE” refers to thermoplastic elastomer.

[0043] In this disclosure, one or more publications, patents, and/or patent applications may be incorporated by reference. However, such material is only incorporated to the extent that no conflict exists between the incorporated material and the statements and drawings set forth herein. In the event of any such conflict, including any conflict in terminology, the present disclosure is controlling.

Overview

[0044] In general, a sole for footwear in accordance with the present disclosure includes a heel portion, a forefoot portion, and a segmentation zone disposed between the heel and forefoot portions. The sole may be configured to be incorporated into an article of footwear, namely a shoe. In some examples, the sole includes a midsole coupled to an outsole such that the outsole is configured to be a ground-contacting surface of the sole and the midsole is configured to be coupled to bottom surfaces of a shoe upper, such as a Strobel board, sockliner, and/or the like. The segmentation zone may include one or more segments or wings, which are separated from each other and from the heel and forefoot portions by a plurality of sipes, cuts, or gaps extending transverse to a longitudinal axis of the shoe.

[0045] The sipes radiate outward from a common origin or originating area disposed along an arch of the sole toward a medial edge of the sole. The common origin (AKA connection point) are disposed on a lateral side of the longitudinal axis. In some examples, the common origin is disposed at a lateral edge of the sole. In some examples, the common origin is disposed slightly interior to the lateral edge and may be separated from the lateral edge by a separation distance. In some examples, the sole includes a notch

disposed between the lateral edge and the common origin, which may add additional flexibility to the segmentation zone.

[0046] The sipes are oriented generally transverse to the length of the sole, and are configured such that the sole is segmented generally orthogonally with respect to a direction of travel. The sipes are configured such that the orthogonal segmentation is disposed in a position roughly corresponding to the arch of the wearer. This configuration of sipes enables bending (i.e., compression and tension) along an area of the sole that is medial to the longitudinal axis of the sole, such that the forefoot and heel of the sole can move in a lateral direction (i.e., sideways) in response to corresponding movement of the human foot.

[0047] The sole includes at least two sipes, each adjacent pair of which may collectively define outer edges of a respective wing-shaped segment extending from the common origin to the medial edge of the sole. In some examples, the sole includes at least three sipes, which define two segments extending from the connection point to the medial edge of the sole. In some examples, the sipes are tapered and substantially triangular, with a sipe width being greater at the medial edge of the sole than at the common origin, facilitating compression in the gaps between the segments and along the longitudinal axis of the shoe.

[0048] In some examples, the segments are substantially pyramidal from medial to lateral edge, having a roughly square base disposed along the medial edge of the sole. Ground-contacting surfaces and top surfaces of the segments may be triangular, and the segments may slope upward from the common origin to the medial edge, providing arch support along top surfaces of the segments. This configuration of sipes facilitates the prevention of overpronation while providing increased flexibility when compared with non-segmented soles.

[0049] In general, an article of footwear or shoe including a sole according to the present disclosure includes an upper bonded to a sole and a flexible insole disposed above the sole and inside an interior region of the article of footwear. In some embodiments, the upper is Strobel-lasted and the flexible insole is sewn to a lower edge of the upper, forming the interior region of the shoe. The sole may then be bonded to bottom surfaces of the upper-insole assembly (e.g., a Strobel board).

[0050] In general, an illustrative method of manufacture for a segmented sole for an article of footwear may include providing a sole, wherein the sole includes a midsole and an outsole, and segmenting the sole to produce a plurality of segments separated by sipes.

Examples, Components, and Alternatives

[0051] The following sections describe selected aspects of exemplary segmented footwear soles as well as related systems and/or methods. The examples in these sections are intended for illustration and should not be interpreted as limiting the scope of the present disclosure. Each section may include one or more distinct embodiments or examples, and/or contextual or related information, function, and/or structure.

A. Illustrative Articles of Footwear

[0052] As shown in FIGS. 1-12, this section describes illustrative articles of footwear including segmented soles.

[0053] As depicted in FIGS. 1 and 2, a first article of footwear or shoe **100** including a segmented sole includes an upper **102** coupled to a sole **110**. Sole **110** includes an outsole **114** coupled (e.g., bonded) to a bottom surface of a midsole **112**. A flexible sockliner or insole **116** is disposed on a top surface of the midsole, or on a top surface of a Strobel board forming a bottom surface of the upper. The shoe may be Strobel lasted (AKA slip, force, or California lasting), wherein a Strobel board is sewn, glued, or otherwise attached to the upper such that the Strobel board forms a bottom surface of an interior region of the shoe. In some examples, the Strobel board comprises sockliner **116**. Upper **102** includes a foot opening **104** disposed at a top anterior surface of the upper. In some embodiments, the upper of the shoe is laced, and includes a shoelace and tongue.

[0054] Upper **102** may include any suitable textile or combination of textiles typically used in shoe manufacture, such as a resilient or elastic material (e.g., nylon, lycra, polyester, polyurethane, and/or the like). In some examples, the sockliner **116** comprises polyurethane, ethylene-vinyl acetate (EVA), and/or another resilient polymer allowing for support and flexibility in the sole of a wearer's foot. In some examples, the sockliner **116** comprises nylon, lycra, and/or the like. The shoe may be configured to allow the arch of a wearer of the shoe to flatten along a longitudinal axis of the shoe. Upper and/or sockliner materials may be selected to stretch and flatten with the foot such that the arch of the wearer can flatten as the foot moves.

[0055] In some examples, a supplemental insole **118** (see FIG. 12) is disposed above integrated sockliner **116** and inside the interior of the shoe, and may provide additional cushioning and support to a foot of the wearer. In some examples, the sockliner and/or supplemental insole of the shoe includes reinforcements to provide additional arch support for a wearer of the shoe, such as resilient polymer inserts, areas of increased thickness around the arch of the shoe, and/or added plates and/or shanks to reduce flexibility.

[0056] In some examples, supplemental insole **118** includes segmentation similar to the sole segmentation described above. (See FIG. 12) Supplemental insole **118** includes one or more sipes **119** disposed transverse to a length of the sole. In some examples, sipes **119** extend from a lateral edge of the insole toward a medial edge of the insole. In some examples, the upper is reinforced with polymers such as polyurethane, EVA, polyester, and/or other polymers configured to resist loads applied by a wearer's weight. The reinforcements may be applied to areas of the upper corresponding to shoelace or segment attachment points.

[0057] With reference to FIG. 2, sole **110** is segmented around a segmentation axis **120** disposed generally orthogonal to a longitudinal axis **122** of the shoe. The sole includes a heel portion **124**, a forefoot portion **126**, and a segmentation zone **128** disposed between the heel and forefoot of the shoe. The segmentation zone includes one or more segments **130**, which are separated from each other and from the heel and forefoot portions by a plurality of grooves, cuts, gaps, or sipes **140** extending transverse to the longitudinal axis **122** of the shoe. In some examples, segments **130** may be substantially triangular. In some examples, the sole includes a minimum of three sipes and two wing segments, facilitating a distribution of pressure from the weight of a wearer along the segmentation zone.

[0058] The number of sipes 140 and wing segments 130 may differ based on one or more functional variables. A sole including a greater number of sipes may be more flexible, while a sole including fewer sipes may be stiffer or more rigid. Thicker midsoles may also utilize a greater number of sipes when compared with thinner midsoles to achieve an approximately equivalent level of sole flexibility. More rigid sole materials may utilize a greater number of sipes when compared with more elastic sole materials to achieve a comparable level of flexibility. Soles may include a greater number of sipes to distribute bending force along a larger region of the shoe, e.g., to increase shoe durability. Soles designed for shoes of larger sizes may include more sipes than smaller shoes.

[0059] Sipes 140 radiate outward from a common region 142, disposed along an arch of the sole, toward a medial edge 144 of the sole. In this example, common region 142 is disposed on a lateral side of the longitudinal axis 122. In some examples, a notch 146 in the sole is disposed at a lateral edge 145 near common region 142. This configuration facilitates increased flexibility within the segmentation zone.

[0060] Sipes 140 may have any suitable widths. Shoes including wider sipes may be more flexible than shoes including narrower sipes, as the sipes act as compression and expansion zones for the shoe. Sipes may also have varying lengths, with shorter sipes providing greater flexibility across the medial side of the arch, and longer sipes providing flexibility along more of a width of the foot. Shoes may include sipes of varying widths and lengths within a single sole. For example, sipes disposed closer to segmentation axis 120 may be shorter than sipes disposed closer to heel and forefoot regions of the shoe.

[0061] Sole 110 has a ground-contacting surface 129, which at heel portion 124 may be substantially flat, allowing for stability while a wearer is standing, and which at forefoot portion 126 may be rounded or tapered, facilitating a rolling movement through the foot while walking or running. The sole may be configured to facilitate a complete rolling through the foot by the wearer, as a function of the increased flexibility of the sole in the segmentation zone.

[0062] FIG. 3 depicts a bending of the sole in normal use. Rolling through the foot flexes the sole throughout segmentation zone 128, increasing a length of the shoe as a length of the wearer's arch increases.

[0063] FIG. 4 depicts a side view of segments 130 included within the article of footwear. The segments are separated from each other and from the forefoot and heel portion by sipes 140, which define outer edges of the segments. In some examples, the segments taper from the lateral edge of the shoe to the medial edge of the shoe, with a triangular profile when viewed from below and a substantially square profile when viewed from the medial side of the shoe. In some examples, segments 130 are substantially pyramidal. Segments 130 are configured to support the arch of a wearer of the shoe, and may include a curved top surface 132. Segments 130 can have different sizes and profiles without departing from the scope of the invention.

[0064] In some examples, segmented sole 110 is coupled to upper 102 of shoe 100 using adhesive, sewing, and/or other suitable attachment methods, e.g., a single-use fastener. In some examples, segmented sole 110 is coupled to a bottom surface of sockliner 116 using adhesive. In some examples, forefoot and heel regions of sole 110 are coupled

to upper 102 using a first attachment method (e.g., adhesive, sewing, etc.) and segments 130 are coupled to upper 102 using a second attachment method (e.g., adhesive, sewing, etc.). In some examples, the wing segments are coupled to the laces of the shoe, which pull upward on the wing segments when tied, thereby providing greater arch support to a wearer of the shoe. In some examples, the wing segments wrap at least partially around the upper of the shoe, and therefore the user's foot, improving shoe durability and providing increased arch support.

[0065] In some implementations, midsole 112 includes two or more midsole layers (e.g., in a dual configuration comprising an external midsole facing the outsole 114, and an internal midsole facing the insole 116). In some examples midsole 112 includes an internal midsole disposed proximate an interior of the shoe and an external midsole disposed proximate ground-contacting surface 129, 129A. In some examples, such as when a Strobel board of the shoe comprises sockliner 116, the Strobel board can be positioned between the external midsole and the external midsole. In some examples, the internal midsole can be disposed within a "sock" of the shoe comprising upper 102 and sockliner 116, which is configured to contain the foot of a wearer. In some examples, sipes 140 are formed only in the external midsole. In other examples, the sipes are formed in the external midsole and at least part of the internal midsole.

[0066] FIGS. 5-8 depict a variety of illustrative midsole-outsole configurations. Different midsole-outsole configurations result in shoes having varying levels of flexibility, traction, and/or durability. FIGS. 5-6 depict a shoe 100A including an upper 102A having a foot opening 104A disposed at a top anterior surface of the upper. Shoe 100A is substantially similar to shoe 100, except as described below. Sole 110A includes an outsole 114A coupled (e.g., bonded) to a bottom surface of a midsole 112A. A flexible sockliner or insole 116A is disposed on a top surface of the midsole, or on a top surface of a Strobel board forming a bottom surface of the upper. As described above, sole 110A is segmented around a segmentation axis 120A disposed generally orthogonal to a longitudinal axis 122A of the shoe. Sole 110A has an outsole 114A which covers a majority of a ground-contacting surface 129A of the shoe. While outsole 114A does not contact wings or sipes of segmentation zone 128A, the sole depicted in FIGS. 5-6 may have decreased flexibility when compared with the sole depicted in FIGS. 7-8.

[0067] FIGS. 7-8 depict a shoe 100B including an upper 102B having a foot opening 104B disposed at a top anterior surface of the upper. Shoe 100B is substantially similar to shoe 100, except as described below. Sole 110B includes an outsole 114B coupled (e.g., bonded) to a bottom surface of a midsole 112B. A flexible sockliner or insole 116B is disposed on a top surface of the midsole, or on a top surface of a Strobel board forming a bottom surface of the upper. As described above, sole 110B is segmented around a segmentation axis 120B disposed generally orthogonal to a longitudinal axis 122B of the shoe. Sole 110B having a more minimal outsole 114B, with the outsole only covering areas of the sole which require greater amounts of grip. Portions of midsole 112B may form portions of a ground-contacting surface 129B of the sole. Flexibility of sole 110B within a segmentation zone 128B may be similar to flexibility of sole 110A within segmentation zone 128A. However, sole 110B may have increased flexibility within a heel region and

forefoot region **124B**, **126B** when compared with sole **110A** due to more flexible midsole materials.

[0068] Midsoles **112A**, **112B** may comprise lightweight materials configured to cushion the sole, such as EVA, polyurethane, TPU, and/or other force-absorbent polymers. Outsoles **114A**, **114B** may comprise any suitable durable, resilient materials configured to provide grip and durability, such as rubber and/or the like. Sole materials may be selected based on a variety of material properties, such as flexibility, durability, weight, and/or other factors affecting shoe performance. The midsole and/or outsole include a tread pattern configured to increase sole traction on underlying surfaces. As typical midsole materials are more lightweight than typical outsole materials, the sole depicted in FIGS. 7-8 may have increased flexibility when compared with the sole depicted in FIGS. 5-6.

[0069] The soles depicted in FIGS. 5-8 include an illustrative common region **142A**, **142B** for use in sole **110A**, **110B**. Common region **142A**, **142B** is recessed when compared with heel region **124A**, **124B** and forefoot portion **126A**, **126B**. As described above, common region **142A**, **142B** may include a notch or gap **146A**, **146B** disposed at a lateral edge **145A**, **145B**, which may increase flexibility within the common region.

[0070] Segments **130A**, **130B** extend from bottom surfaces (when in use) of common region **142A**, **142B**, toward a medial edge **144A**, **144B** of the sole, and have an increased thickness when compared with the common region. In some examples, ground-contacting surfaces of segments **130A**, **130B** are substantially coplanar with ground-contacting surfaces of the heel and forefoot regions. This configuration may further increase flexibility of the sole, as recess **143A**, **143B** below common region **142A**, **142B** may function similarly to sipes **140A**, **140B** when the shoe is in use, compressing and extending along longitudinal axis **122A**, **122B**.

[0071] FIGS. 9-11 depict a shoe **200** having a segmented sole **210**, including a forefoot portion **226**, a heel portion **224**, and a segmentation zone **228** disposed between the forefoot portion and the heel portion. Segmentation zone **228** includes a single notch or gap **240** between the forefoot portion and the heel portion, with a forefoot edge **244** and a heel edge **246** extending radially outward from a common region **242** disposed along an area of the sole corresponding to an arch of a wearer's foot. An elastic material may extend between the forefoot edge and heel edge, providing a support component **248** for the arch of the wearer. The shoe may otherwise be substantially identical to the shoe described above with respect to FIGS. 1-8.

[0072] The following reference numerals are utilized in the accompanying drawings:

- [0073] **100**, **100A**, **100B** shoe
- [0074] **102**, **102A**, **102B** upper
- [0075] **104**, **104A**, **104B** foot opening
- [0076] **110**, **110A**, **110B** sole
- [0077] **112**, **112A**, **112B** midsole
- [0078] **114**, **114A**, **114B** outsole
- [0079] **116**, **116A**, **116B** sockliner
- [0080] **118** supplemental insole
- [0081] **119** insole sipes
- [0082] **120**, **120A**, **120B** segmentation axis
- [0083] **122**, **122A**, **122B** longitudinal axis
- [0084] **124**, **124A**, **124B** heel portion
- [0085] **126**, **126A**, **126B** forefoot portion

- [0086] **128**, **128A**, **128B** segmentation zone
- [0087] **129**, **129A** ground-contacting surface
- [0088] **130**, **130A**, **130B** segments
- [0089] **132** curved top surface
- [0090] **140**, **140A**, **140B** sipes
- [0091] **142**, **142A**, **142B** common region
- [0092] **143A**, **143B** recess
- [0093] **144**, **144A**, **144B** medial edge
- [0094] **145**, **145A**, **145B** lateral edge
- [0095] **146**, **146A**, **146B** notch
- [0096] **200** shoe
- [0097] **210** segmented sole
- [0098] **224** heel portion
- [0099] **226** forefoot portion
- [0100] **228** segmentation zone
- [0101] **240** notch or gap
- [0102] **242** common region
- [0103] **244** forefoot edge
- [0104] **246** heel edge
- [0105] **248** support component

B. Illustrative Method of Manufacture

[0106] This section describes steps of an illustrative method **1000** for manufacturing segmented footwear soles. Aspects of illustrative articles of footwear and illustrative soles, described above, may be utilized in the method steps described below. Where appropriate, reference may be made to components and systems that may be used in carrying out each step. These references are for illustration, and are not intended to limit the possible ways of carrying out any particular step of the method.

[0107] FIG. 13 is a flowchart illustrating steps performed in an illustrative method, and may not recite the complete process or all steps of the method. Although various steps of method **1000** are described below and depicted in FIG. 13, the steps need not necessarily all be performed, and in some cases may be performed simultaneously or in a different order than the order shown.

[0108] Step **1002** of method **1000** includes providing a sole for an article of footwear (e.g., an athletic shoe), the sole including a midsole coupled to an outsole. In some embodiments, providing a sole includes procuring a premanufactured sole from another party. In some examples, providing a sole includes manufacturing a sole for an athletic shoe using thermoforming, injection molding, 3D printing, and/or any suitable method for producing a foam or polymer sole for an athletic shoe. In some examples, providing a sole includes providing a premanufactured midsole and a premanufactured outsole and subsequently bonding the midsole and outsole using adhesive, mechanical attachment methods, and/or other suitable methods for bonding two sole components. In some examples, the midsole and/or outsole may comprise any of a variety of polymers, including EVA, PU, TPU, Nylon, rubber, TPE, and/or any other suitable polymer for producing footwear soles. In some examples, the midsole may comprise PU, EVA, TPU, nylon, and/or any other polymer with cushioning or energy returning properties. In some examples, the outsole may comprise durable polymers such as rubber, TPE, and/or any other polymer with advantageous grip and wear properties.

[0109] Step **1004** includes segmenting the sole by cutting or otherwise creating sipes. In some examples, each sipe may include a single cut disposed transverse to a longitudinal axis of the sole. In some examples, each sipe may

include two cuts disposed transverse to the longitudinal axis, such that the sipes are tapered, e.g., widest at a medial edge of the shoe and decreasing in width as they extend toward a lateral side of the shoe.

[0110] In examples where the sipes are cut into the sole (e.g., rather than being molded), cutting sipes into the sole may comprise any suitable method for cutting a preformed polymer or foam component. For example, this may include die cutting, laser cutting, cutting with a water jet cutter, and/or any other suitable method for cutting shapes from a prefabricated polymer component. Cutting the sipes may be done by hand or may be done using CNC (computer numeric control) or other forms of automated cutting.

[0111] In some examples, steps 1002 and 1004 are performed simultaneously. In these examples, thermoforming or injection molding the sole may include providing a mold including premanufactured sipe-forming regions. These regions may comprise appendages, castellations, and/or other protrusions included within a mold which prevent deposition of material in certain areas, producing sipes or gaps or cuts within the sole.

[0112] Step 1004 may optionally be followed by step 1006, which includes coupling the completed sole to a pre-manufactured upper to produce a completed shoe or other article of footwear. The completed sole may be fastened by bonding to the pre-manufactured upper using adhesive and/or other suitable mechanical attachment methods.

C. Illustrative Method of Use

[0113] An illustrative method of use for an article of footwear or shoe including a segmented sole includes flexing the sole along a direction of anatomical movement in a foot wearing the shoe. This direction of anatomical movement corresponds to a longitudinal axis of the shoe, extending between the forefoot portion of the shoe and the heel of the shoe. Flexing the sole in a first direction, such that the heel and forefoot are caused to move toward the top of the shoe, causes the sipes to expand in width, lengthening the shoe. The upper stretches with the sole, and the arch of a wearer stretches with the sole. Flexing the sole in a second direction, such that the heel and forefoot are moved toward the bottom of the shoe, causes the sipes to compress in width, contracting the shoe. This may be useful for dance applications or other activities such as hiking or trail running which require a high degree of maneuverability. Applying uniform downward pressure across the shoe, such as when standing, also causes the arch of a wearer to flatten, as the (e.g., elastic) upper facilitates a stretching of the foot along a longitudinal axis.

D. Illustrative Combinations and Additional Examples

[0114] This section describes additional aspects and features of segmented footwear soles, presented without limitation as a series of paragraphs, some or all of which may be alphanumerically designated for clarity and efficiency. Each of these paragraphs can be combined with one or more other paragraphs, and/or with disclosure from elsewhere in this application, including the materials incorporated by reference in the Cross-References, in any suitable manner. Some of the paragraphs below expressly refer to and further limit

other paragraphs, providing without limitation examples of some of the suitable combinations.

[0115] A0. A sole for an article of footwear, comprising: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a plurality of segments separated from each other by one or more sipes; wherein the segmentation zone is configured such that the sole compresses and expands as a function of compression and expansion within the sipes.

[0116] A1. The sole of paragraph A0, wherein the plurality of sipes includes at least three sipes.

[0117] A2. The sole of any one of paragraphs A0-A1, wherein each of the plurality of segments is substantially triangular in shape.

[0118] A3. The sole of any one of paragraphs A0-A2, wherein the sole comprises a midsole and an outsole.

[0119] A4. The sole of paragraph A2, wherein the midsole comprises polyurethane.

[0120] A5. The sole of any one of paragraphs A3-A4, wherein the outsole comprises rubber.

[0121] B0. An article of footwear, comprising: an upper including a resilient sockliner; and a sole, the sole including: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a plurality of segments separated from each other and from the heel and forefoot portions by a plurality of sipes; wherein the segmentation zone is configured such that the sole is bendable as a function of compression within the sipes.

[0122] B1. The article of footwear of paragraph B0, wherein the plurality of sipes includes at least three sipes.

[0123] B2. The article of footwear of any one of paragraphs B0-B1, wherein each wing of the plurality of segments is substantially triangular in shape.

[0124] B3. The article of footwear of any one of paragraphs B0-B2, wherein the sole comprises a midsole and an outsole.

[0125] B4. The article of footwear of paragraph B2, wherein the midsole comprises polyurethane.

[0126] B5. The article of footwear of any one of paragraphs B3-B4, wherein the outsole comprises rubber.

[0127] B6. The article of footwear of any one of paragraphs B0-B5, wherein the upper comprises nylon.

[0128] B7. The article of footwear of any one of paragraphs B0-B6, wherein the midsole is bonded to the upper using adhesive.

[0129] B8. The article of footwear of any one of paragraphs B0-B7, wherein the article of footwear comprises a shoelace.

[0130] B9. The article of footwear of paragraph B8, wherein each segment of the plurality of segments is coupled to the upper by the shoelace.

[0131] C0. A method for manufacturing a sole for an article of footwear, comprising: providing a sole; and adding a plurality of sipes to the provided sole; wherein adding sipes to the provided sole produces a segmentation zone, the segmentation zone including a plurality of segments, wherein the segments are separated from each other and from the heel and forefoot portions by the plurality of sipes.

- [0132] C1. The method of paragraph C0, wherein adding the plurality of sipes to the provided sole includes die cutting.
- [0133] C2. The method of paragraph C0, wherein adding the plurality of sipes to the provided sole includes injection molding.
- [0134] C3. The method of paragraphs C0-C2, wherein the plurality of sipes includes at least three sipes.
- [0135] D0. The construction of a shoe wherein the midsole and outsole are segmented and/or split along an orthogonal axis to a perpendicular vector in which anatomical movement would occur on the human foot.
- [0136] D1. This segmentation can be accomplished via, but is not limited to: breaking, material combinations, filler materials with engineered elastic modulus, top-down gapping, and bottom-up gapping.
- [0137] D2. The segmentation along given axis is such that a moment arm correspondent to the axial break or opposite the midsole, outsole, or a combination of the two, allows for both a tension and compression force that aligns with movement of the human foot.
- [0138] E0. A product comprising any feature described herein, either individually or in combination with any other such feature, in any configuration.
- [0139] F0. A process for manufacturing a sole for an article of footwear, the process comprising any process step described herein, in any order, using any modality.

Advantages, Features, and Benefits

- [0140] The different embodiments and examples of the segmented footwear sole described herein provide several advantages over known solutions for providing flexible, but supportive shoe soles. For example, illustrative embodiments and examples described herein allow a wearer of the shoe to experience shoe bending which allows their foot to move in a natural anatomical motion.
- [0141] Additionally, and among other benefits, the footwear sole described herein allows the arch of a wearer to flatten during use.
- [0142] Additionally, and among other benefits, the footwear sole described herein supports the arch of a wearer.
- [0143] No known system or device can perform these functions. However, not all embodiments and examples described herein provide the same advantages or the same degree of advantage.

CONCLUSION

[0144] The disclosure set forth above may encompass multiple distinct examples with independent utility. Although each of these has been disclosed in its preferred form(s), the specific embodiments thereof as disclosed and illustrated herein are not to be considered in a limiting sense, because numerous variations are possible. To the extent that section headings are used within this disclosure, such headings are for organizational purposes only. The subject matter of the disclosure includes all novel and nonobvious combinations and subcombinations of the various elements, features, functions, and/or properties disclosed herein. The following claims particularly point out certain combinations and subcombinations regarded as novel and nonobvious. Other combinations and subcombinations of features, functions, elements, and/or properties may be claimed in applications claiming priority from this or a related application.

Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

What is claimed is:

1. A sole for an article of footwear, the sole comprising: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a first sipe on a medial edge of the sole and a second sipe on the medial edge of the sole, wherein the first sipe and the second sipe angle toward each other, intersect at a convergence point inset from a lateral edge of the sole, and combine into a combined sipe extending between the convergence point and the lateral edge of the sole.
2. The sole of claim 1, wherein the first sipe and the second sipe define a triangular segment between the medial edge of the sole and the convergence point.
3. The sole of claim 2, wherein a side of the triangular segment is along the medial edge of the sole and an apex of the triangular segment is at the convergence point.
4. The sole of claim 3, wherein the sole includes a third sipe extending into and partially dividing the triangular segment.
5. The sole of claim 4, wherein the third sipe extends from the side of the triangular segment.
6. The sole of claim 5, wherein the third sipe does not connect to the convergence point.
7. The sole of claim 1, wherein a width of the combined sipe is greater than a width of the first sipe or is greater than a width of the second sipe.
8. A sole for an article of footwear, the sole comprising: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone including a first sipe extending between a medial edge of the sole and a lateral edge of the sole and a second sipe extending between the medial edge of the sole and the lateral edge of the sole, wherein the first sipe and the second sipe intersect at a convergence point inset from the lateral edge of the sole and define a triangular segment between the medial edge of the sole and the convergence point.
9. The sole of claim 8, wherein a side of the triangular segment is along the medial edge of the sole and an apex of the triangular segment is at the convergence point.
10. The sole of claim 8, wherein the sole includes a third sipe extending from the medial edge toward the convergence point.
11. The sole of claim 10, wherein the third sipe extends from a side of the triangular segment.
12. The sole of claim 10, wherein the third sipe extends into and partially divides the triangular segment.
13. The sole of claim 10, wherein the third sipe is opposite the convergence point.
14. The sole of claim 8, wherein the lateral edge of the sole curves inward opposite the convergence point.
15. A sole for an article of footwear, the sole comprising: a heel portion; a forefoot portion; and a segmentation zone disposed between the heel portion and the forefoot portion, the segmentation zone includ-

ing a first sipe extending between a medial edge of the sole and a lateral edge of the sole, a second sipe extending between the medial edge of the sole and the lateral edge of the sole, and a third sipe, wherein the first sipe and the second sipe meet defining a segment between the medial edge and the lateral edge, and the third sipe extends into and partially divides the segment.

16. The sole of claim **15**, wherein the third sipe extends from the medial edge of the sole.

17. The sole of claim **15**, wherein the segment is triangular.

18. The sole of claim **15**, wherein a side of the segment is along the medial edge of the sole.

19. The sole of claim **15**, wherein the lateral edge of the sole curves inward opposite an apex of the segment.

20. The sole of claim **15**, wherein the medial edge curves inward along an edge of the segment.

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